Dilmin Siriwardana

OBJECTIVE

Passionate Mechatronics Engineering graduate with hands-on experience in various programming languages, CAD design, and PCB designing. Specializing in innovative engineering solutions across multiple industries with current focus on Machine Learning, Artificial Intelligence and Robotic Operating System applications. Proven track record in team collaboration and project management. Demonstrated leadership abilities through executive roles in student organizations and strong commitment to continuous learning and professional development.

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WEBSITE

https://dilmins.github.io/

LINKEDLN

https://www.linkedin.com/in/dilmin-siriwardana

EXPERIENCE

Mechatronic Engineer | C3 Labs Pvt Ltd | Colombo 04

MAR 2026 – Present

- Developing firmware's for E-paper displays with cloud-connected IoT platforms, combining hardware and software aspects.
- Researching and developing ultra-low battery usage firmware's between 2 microcontrollers with various communication protocols
- Conducting device testing, fault diagnosis, and calibration to ensure optimal performance and safety standards.

Research And Development Intern Engineer | C3 Labs Pvt Ltd | Colombo 04

OCT 2024 – MAR 2025

- Developed and integrated E-paper display systems with cloud-connected IoT platforms, combining hardware and software aspects.
- Implemented hardware–software interfacing for data transmission and device communication protocols.
- Assisted in PCB circuit design and component optimization, improving power efficiency and device reliability.
- Conducted system testing, fault diagnosis, and calibration to ensure optimal performance and safety standards.
- Provided technical support to external operators for installation, setup, and troubleshooting of devices.
- Mentored junior interns, guiding them through circuit debugging and integration practices.

Engineering Intern | Vidullanka PLC | Colombo 03

JUN 2021- NOV 2021

- Collaborated with the Engineering Department to implement IoT-based level monitoring units across 9 hydro power plants, using ultrasonic and sensor-based solar-powered units at forebays and spillways.
- Gained hands-on experience with flow measurement using automatic gauge units and learned to identify critical operational aspects of each hydro power plant.
- Supported the HR Department during the COVID period by managing documents, conducting online training sessions for remote staff to enhance MS Office skills, and helping implement Japanese techniques such as 5S, 3R, and KAIZEN.
- Observed and learned how meetings are conducted to track progress, monitor work performance, and guide staff toward achieving organizational goals.
- Participated in organizing employee engagement events such as award ceremonies, cricket tournaments, and family trips to strengthen team spirit.

SOFT SKILLS

- Team Work
- Problem-Solving & Critical Thinking
- Time Management & Multitasking
- Adaptability & Quick Learning
- Communication Skills
- Familiarity with 4DX (The 4 Disciplines of Execution)

PROFESSIONAL DEVELOPMENT

Participating in a Leadership for Business Growth training program conducted by JASTECA & AOTS Tokyo, Japan

- Focused on ESG, Big Data, and Japanese Management Techniques (Servicology and Inclusive Leadership).
- Conducted industrial study tours at Mazda Motor Corporation, AIST, and Yamako Co., Ltd.

REFERENCES

- Mr. Shehan Priyanga Fernando Bsc. Eng. (Moratuwa), M.Sc. In Financial Mathematics (Moratuwa), M.Phil. (Moratuwa) Senior Lecturer Gr I General Sir John Kotelawala Defence University Tel : 071 832 3855

- Mr. Tony Mahadevan Bsc. It. (Manchester Met) Chief Executive Officer C3 Labs Pvt Ltd Tel : 0777312881

EDUCATION

- **Bachelor of Mechatronic Engineering (Pending)** 2022 - 2026
  General Sir John Kotelawala Defence University
- **CIMA – Certificate Level (Ongoing)** 2026
  Wisdom Business Academy
- **Certificate in Automation (PLC)** 2024
  Sri Lanka Institute of Robotics
- **Diploma in Office Applications** 2022
  IDM Achievers (Gold Award Winner)
- **Cambridge A/Levels & O/Levels** 2005 - 2021
  Lyceum International School Panadura

TECHNICAL SKILLS

Programming Languages: Python, C++, PLC Ladder Programming

Design & CAD: PCB Designing , Mechanical Drawings

Software & Tools: Altium Designer, MATLAB, Arduino IDE, VS Code, Postman, Fritzing, MS Office, SIMATIC Manager, TIA Portal, Step-7 Micro WIN

Specialized Areas: Embedded Systems and IoT Backend Programming , AI/Machine Learning/Deep Learning, Automation

KEY PROJECTS

1. **Brain-Controlled Exoskeleton Arm for Assisting Hand-Impaired Individuals using EEG Signals** (2025 - Ongoing)
   - Tools Used : MATLAB, EEGLab , Python , C++ , SolidWorks, Raspberry Pi4
   - Associated with : General Sir John Kotelawala Defence University
   - Final Demonstration Video - https://www.linkedin.com/posts/dilmin-siriwardana-final-year-reasearch-project-brain-controlled-activity-7407598097700130817OJGF?utm_source=share&utm_medium=member_desktop&rcm=ACoAADZJzQBSRr6Qkp6u5m8NzCtWiQd6rWR78E
2. **Designing of a PCB Power Triggering Circuit using NFC Method** - (2025)
   - Tools Used : Altium Designer
   - Associated with : C3 Labs Pvt Ltd
3. **Programming of QR Code Scanning Device for Transactions** - (2025)
   - Tools Used : Arduino IDE, The Dot Factory, Postman, ESP32
   - Associated with : C3 Labs Pvt Ltd
4. **Programming of Air Quality Monitoring (AQM) Devices** - (2024)
   - Tools Used : Arduino IDE , The Dot Factory, Postman, ESP32
   - Associated with : C3 Labs Pvt Ltd
5. **ML based Surveillance Robot Car** - (2024)
   - Tools Used : OpenCV, VS Code, Arduino IDE, Raspberry Pi4,
   - Associated with : General Sir John Kotelawala Defence University
6. **Automated Cocktail Machine** - (2024)
   - Tools Used : OpenCV, Tkinter, VS Code, Arduino IDE, Raspberry Pi4
   - Associated with : General Sir John Kotelawala Defence University

EXTRA CURRICULAR ACTIVITIES

- Executive Committee Member 2023- 2024
  IMECHE - General Sir John Kotelawala Defence University
- Director of Community Services 2023- 2024
  Rotaract Club of Centennial United
- Member 2022- 2023
  Rotaract Club of General Sir John Kotelawala Defence University
- Member since 2026
  AOTS & JASTECA